GROUNDCONTROL NODAL SERIES

Tuned Diaphragmatic Absorber (TDA)

Design & Physics White Paper

Acoustic Modal Simulation Model — r61

Physics basis: Mechel · Cox & D'Antonio · Morse & Ingard · Kuttruff · Eyring

Indicative predictions, validate all results with REW measurement · not for engineering certification

1. Executive Summary

This white paper documents the physics model underpinning the GroundControl Nodal Series Tuned Diaphragmatic Absorber (TDA). The Nodal Series is a low-frequency room treatment product line designed to address axial and near-axial modal resonances in small rooms — the primary source of coloration, uneven bass response, and excessive modal decay in recording studios, mastering suites, home theatres, and broadcast control rooms.

The model is implemented in the GroundControl Acoustic Modal Simulation tool (r58). It predicts pressure distribution, dB modal reduction, and per-mode decay time (T60) as a function of unit tuning frequency, quality factor, placement, room geometry, and construction profile. Results are validated against published impedance-tube and reverberation-room data and calibrated against REW measurements. This document reflects model version r61. Key r61 additions: Nodal 40 renamed to Nodal 30 (35 Hz Q=1.0); Nodal 50 retired — replaced by collapsible Advanced Custom Unit 2 defaulting to Nodal 30 parameters; broadband panel coverage % and depth selector (2″/4″/6″) with dynamic OC703 ISO 354 alpha; Treatment Progress replaced by Treatment Recommendation panel; modalCouplingArea corrected to use A_term only (fixing TDA δ underestimation by ~2.8×); PDF report expanded with broadband panel contribution table. From r60: REW waterfall T60M import replacing RT Parameters T30 import — the model now computes T60M at all room mode frequencies directly from REW waterfall exports using a Nelder-Mead STFT fitting engine; continuous measured T60 curve across 20–150 Hz in the Bass Modal Decay graph; delta_boundary corrected to use full room surface area (Sabine-weighted) rather than modal coupling area only, fixing a systematic T60 Pre over-prediction of 3–4× in residential rooms; default Nodal 30 target updated to 35 Hz Q=1.0; default cavity depth updated to 215 mm; Rise Time default updated to 30 ms. Earlier additions: Treatment Recommendation panel (r61, replaces Treatment Progress), listening position analysis (r58), URL config save/load (r58), revised AVAA confidence band (r58), REW waterfall import framework (r59), sloped ceiling perturbation (r55), cathedral ceiling (r56), user-adjustable membrane dimensions (r57).

Key outputs of the model:

- Per-mode T60 before and after treatment, with saturation-aware predicted measurement range
- Broadband RT60 across 20–150 Hz via the Eyring equation
- dB modal pressure reduction displayed as a room heatmap
- Optimal OC 705 fill position per unit (false-back depth recommendation)
- Corner stacking efficiency and unit count recommendations

2. The Low-Frequency Modal Problem

2.1 Axial Room Modes

In a rectangular room of dimensions L × W × H (meters), standing waves (modes) form at integer multiples of the half-wavelength along each axis. The resonant frequencies of axial modes are given by:

$$f(n, d) = n \times c / (2d) \qquad n = 1, 2, 3, \ldots$$

where c = 343 m/s (speed of sound at 20 °C) and d is the room dimension along the relevant axis. The default room in the simulation is 6.1 m × 4.9 m × 2.4 m (20 × 16 × 8 ft), giving the following first-order axial modes:

Axis	Dimension	1st Mode (Hz)	2nd Mode (Hz)
X (Length)	6.1 m / 20 ft	28.1 Hz	56.2 Hz
Y (Width)	4.9 m / 16 ft	35.0 Hz	70.0 Hz
Z (Height)	2.4 m / 8 ft	71.5 Hz	143.0 Hz

Table 1 — First two axial modes for the reference room (6.1 × 4.9 × 2.4 m)

These modes dominate the sound field below the Schroeder frequency, approximately 80–120 Hz for typical small rooms. At modal resonances, SPL can vary by 20 dB or more across the room, and decay times (T60) can reach 2–4 seconds — far longer than the 0.25–0.55 second target range for professional monitoring environments.

2.2 Why Conventional Treatment Fails Below 80 Hz

Porous absorbers (rockwool, OC 703/705 panels) are velocity-based: they dissipate energy through viscous drag in air moving through the material. Effectiveness requires panel thickness comparable to a quarter-wavelength. At 40 Hz, quarter-wavelength is 2.1m which is impractical for any room treatment product.

Conventional broadband panels (4" OC 703) provide meaningful absorption from ~63 Hz upward ($\alpha \approx 0.43$ at 63 Hz, rising to 0.97 at 125 Hz) but contribute negligible energy extraction at 28–55 Hz where the most problematic axial modes typically reside.

The TDA addresses this by operating at modal pressure maxima, the room boundaries, where acoustic pressure (not velocity) is maximum, and using mechanical resonance to extract and dissipate modal energy.

3. TDA Operating Principle

3.1 Piston Resonator Model

A Tuned Diaphragmatic Absorber (TDA) — also called a membrane absorber or panel absorber — consists of a compliant panel (the membrane) suspended over an air cavity. The system behaves as a mass-spring resonator:

- Mass (m): panel surface mass in kg/m²
- Spring (k): adiabatic stiffness of the enclosed air cavity
- Damping (R): resistive material (OC 705 compressed fibreglass) placed within the cavity

The resonant tuning frequency is:

$$f_0 = (1/2\pi) \times \sqrt{(\rho c^2 / (m \times D))}$$

where ρ = 1.21 kg/m³ (air density), c = 343 m/s, m = panel surface mass (kg/m²), and D = active cavity depth (m). Rearranging to solve for required mass:

$$m = (60/f_0)^2 / D$$

Derivation of the constant 60: starting from the tuning equation and substituting air properties (ρ = 1.21 kg/m³, c = 343 m/s):

$$f_0^2 = (c/2\pi)^2 \times \rho / (m \cdot D) \quad \rightarrow \quad m = (c/2\pi)^2 \times \rho / (f_0^2 \cdot D)$$

$$(c/2\pi)^2 \times \rho = (343/2\pi)^2 \times 1.21 = 2980 \times 1.21 \approx 3600$$

$$m = 3600 / (f_0^2 \cdot D) = (60/f_0)^2 / D \qquad [\text{since } \sqrt{3600} = 60]$$

The constant 60 emerges directly from c/(2π) × √ρ ≈ 60 for standard air at 20 °C. It is not an empirical fit.

The Nodal Series uses Richlite (recycled paper composite) as its membrane substrate, with a minimum surface mass of 8.89 kg/m². For tuning frequencies where the required mass exceeds this minimum (approximately below 57 Hz with a 215 mm cavity), Mass Loaded Vinyl (MLV) is added. For higher target frequencies, the cavity depth is reduced to achieve the correct tuning with Richlite alone.

3.2 Unit Specifications

Parameter	Value
Active face area	0.4645 m² (24" × 30" membrane)
Membrane substrate	Richlite 1/4" — surface mass 8.89 kg/m²
Additional mass	MLV (Mass Loaded Vinyl) for f_0 below ~50 Hz
Standard cavity depth	215 mm active (nominal 229 mm outside — 12 mm rear panel, 6 mm Richlite face)
Damping material	Owens Corning 705 compressed fiberglass, position-tunable (false-back depth)
Nodal 30 target	~35 Hz, Q = 1.0
Nodal Custom target (optional — Custom Unit 2)	~35 Hz, Q = 1.0 (Default setting, not used)
Nodal Custom target (optional — Custom Unit 3)	~35 Hz, Q = 1.0 (Default setting, not used)

Table 2 — GroundControl Nodal Series unit specifications

4. Absorption Coefficient Model

4.1 Peak Alpha — Impedance Match (Mechel / Cox & D'Antonio)

The peak absorption coefficient of a wall-mounted panel absorber is not a fixed material property — it depends critically on the ratio of radiation loss to damping loss. The model uses the rigorous impedance-match formulation from Mechel ("Formulas of Acoustics", Ch. 11) and Cox & D'Antonio ("Acoustic Absorbers and Diffusers", 3rd ed., §6.3):

$$\alpha\_peak = \alpha\_max \times 0.5 \times 4R / (1 + R)^2$$

where:

- δ_rad = ρc / (m·ω_0) — radiation loss factor, fixed by the panel mass and resonant frequency
- δ_damp = 1/(2Q) − δ_rad — damping loss factor, controlled by OC 705 fill depth
- R = δ_rad / δ_damp — impedance ratio
- α_max = 0.95 — practical upper limit accounting for panel imperfections
- Factor 0.5: wall-mounted panel radiates from one face only (Mechel §11)

The 4R/(1+R)² term is maximized at R = 1 (impedance match), giving α_peak = α_max × 0.5. This occurs when δ_damp = δ_rad, which defines the optimal Q:

$$Q\_opt = 1 / (4 \cdot \delta\_rad) = m \cdot \omega_0 / (4 \cdot \rho c)$$

Above this Q (less damping than optimal), the panel under-couples to the room, it rings freely but extracts less energy. Below this Q, the panel is over-damped and absorption falls. The model exposes Q as a user parameter to allow exploration of this tradeoff.

Note: *The older exponential saturation formula (α_peak ∝ 1 − exp(−Q/Q_ref)) is replaced by this rigorous impedance model, which correctly predicts the falling α_peak at both high Q (under-damped) and low Q (over-damped) ends.*

4.2 Frequency Response — Skewed Lorentzian

Away from resonance, absorption falls off following a modified Lorentzian (resonance curve). Real panel absorbers exhibit an asymmetric rolloff, steeper below resonance (stiffness-controlled) than above (mass-controlled). The model captures this with a skew correction:

$$x = f/f_0 - f_0/f \qquad \text{(symmetric Lorentzian variable)}$$

$$x\_skewed = x \times (1 + k \cdot x) \qquad k = 0.18$$

$$\alpha(f) = \alpha\_peak / (1 + (Q\_eff \cdot x\_skewed)^2)$$

The skew factor k = 0.18 is calibrated from published panel absorber measurements (Cox & D'Antonio Ch. 6, Fuchs & Aretz, Acustica 1999). This correction is modest (±15% on bandwidth edges) but improves accuracy when comparing predicted and measured curves in the Q = 2–5 operating range.

4.3 Over-Damped Q Correction

For Q < 1.0 (the resistive-dominant regime), the standard Lorentzian overestimates off-resonance effectiveness because a non-resonant resistive panel has a much flatter, lower-amplitude response. An effective Q is computed:

```
Q_eff = Q × exp(k_bw × (Q_floor − Q))    for Q < Q_floor = 1.0
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where k_bw = 1.8 is an empirical calibration parameter, not a cited formula. Cox & D'Antonio §8.4 discusses over-damped panel behavior qualitatively but does not provide this coefficient. The value is calibrated against published panel absorber measurements and should be treated as provisional, particularly for Q < 0.8. Validation against impedance-tube measured panel data is recommended.

- Q = 1.0 → Q_eff = 1.0 (no correction at floor)
- Q = 0.8 → Q_eff ≈ 1.11 (effective bandwidth narrows slightly)
- Q = 0.6 → Q_eff ≈ 1.35 (significant narrowing)
- Q = 0.5 → Q_eff ≈ 1.65 (near-resistive panel, severely narrowed)

A narrower effective bandwidth means off-resonance modes benefit much less from the over-damped unit, matching observed behavior. Note that α_peak is already reduced by the impedance model (R << 1 when over-damped), so no additional peak correction is applied here.

5. OC 705 Placement and False-Back Depth

The quantity of resistive fill (OC 705) and its position within the cavity directly controls Q. The model computes the recommended OC 705 position as a false-back depth, measured from the rear panel toward the membrane, using the impedance-match condition δ_damp = δ_rad.

For each unit, the optimal fill depth is determined by solving:

$$Q\_opt = m \cdot \omega_0 \ / \ (4 \cdot \rho c)$$

and mapping Q to an OC 705 thickness. Shallower fill (less material near the membrane) increases Q (sharper, more selective absorption). Deeper fill (or denser packing) reduces Q toward the resistive-dominant regime.

In practice:

- Q ≈ 1.5–3.0: sharp, frequency-selective. Best for targeting a single isolated mode.
- Q ≈ 0.8–1.2: broad coverage. Useful when two modes fall within $\pm f_0/Q$ of the tuning frequency.
- Q < 0.8: resistive-dominant. The panel still absorbs but is best modelled as a resistive panel, not a true resonator. The model flags this condition.

6. Room Placement and Modal Coupling

6.1 Pressure Maxima

Modal pressure amplitude follows a cosine distribution along each axis. For an axial mode of order n along axis X:

$$P(x) = P\_max \times \cos(n\pi x/L)$$

Pressure maxima occur at the room boundaries: x = 0 (front wall) and x = L (rear wall). All axial and tangential modes share pressure maxima at room corners. This makes corners the optimal placement for TDA units, as all modes are present there, and the pressure driving force is at its maximum amplitude.

6.2 Boundary Pressure Gain

A panel placed at a room boundary experiences a standing-wave pressure doubling relative to the free-field amplitude. In the power domain this becomes a factor of 4× pressure gain (the acoustic pressure squared). At a room corner, two or three boundaries intersect, compounding the gain further.

The model expresses this as geometric gain factors:

- G_corner = 4.0 (default): corner placement. Physically represents the combined pressure doubling from two converging standing waves. From r51, G_corner is automatically calibrated by the REW waterfall import (Section 12): the solver minimizes mean-squared log error between predicted and measured T60M at axial mode frequencies to find the room-specific optimal G. The slider remains available for manual adjustment.
- G_wall = 2.0: wall mid-point placement. Pressure doubling from one boundary only.

6.3 Six-Surface Modal Coupling Area

A key correction versus simpler models is the treatment of modal coupling area. All six room surfaces couple to a mode, but with different pressure amplitudes depending on orientation:

- Terminating walls (perpendicular to the mode axis): pressure maximum; coupling weight = 1.0
- Parallel surfaces (four surfaces parallel to the mode axis): spatial average of $\cos^2(n\pi x/L) = 0.5$ for all $n \geq 1$; coupling weight = 0.5

The effective modal coupling area is:

```
A_eff(axis) = A_term × 1.0 + A_para × 0.5
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where A_term is the area of the two terminating walls and A_para is the area of the four parallel surfaces. This normalization is applied consistently in both the dB modal reduction (heatmap) and the per-mode T60 calculation. Using only the terminating walls (as in earlier model versions) underestimates boundary damping and over-predicts TDA contribution.

Physical basis: Morse & Ingard "Theoretical Acoustics" §9.4; Kuttruff "Room Acoustics" §3.3 modal damping.

6.4 Per-Position Coupling Weight

Beyond the global coupling area, the model assigns each placement position a coupling weight based on its relationship to a given mode:

- Corner positions: effective coupling = 0.75 (average of terminating-face coupling 1.0 and parallel-face coupling 0.5, reflecting the two-column geometry at a floor-plan corner)
- Terminating wall mid-points (WL, WW for X-modes; WR, WF for Y-modes): coupling = 1.0
- Parallel wall mid-points: coupling = 0.5

The total effective absorption contribution per unit group is:

$$A\_i = \alpha(f) \times A\_unit \times G\_pos \times coupling\_i \; / \; A\_eff(axis)$$

6.5 Corner Stacking and Efficiency

Multiple TDA units can be stacked vertically at the same floor-plan corner position. At 20–70 Hz, inter-unit spacing of ~762 mm (30") is a small fraction of the 4.9–17.2 m wavelengths, placing units well within each other's near-field radiation zone. Mutual radiation coupling between adjacent membranes reduces each unit's effective contribution sub-linearly.

Efficiency factors per unit in a stack are calibrated from Fuchs & Zha (2006) and Mechel §12.3. Important limitation: Fuchs & Zha studied acrylic-glass panels at 200–500 Hz, and no published study directly validates these factors for LF diaphragmatic absorbers in small rooms. At 30 Hz (λ ≈ 11.4 m), unit spacing of 0.76 m is λ/15, placing units deep in the near field where coupling may be stronger than these values suggest. Treat as first-order calibration parameters, not derived constants:

Position in Stack	Efficiency Factor	Reduction vs. First Unit
1st unit	1.00	—
2nd unit	0.90	−10%
3rd unit	0.82	−18%
4th unit	0.75	−25%

Table 3 — Stacking efficiency factors. First-order calibration parameters — require per-installation REW T60M waterfall validation. Not validated for LF diaphragmatic absorbers below 100 Hz.

Note: *Fuchs & Zha (2006) source data are for acrylic-glass panels at 200–500 Hz. The application to LF TDA arrays is an extrapolation requiring measurement confirmation.*

Maximum stack height is constrained by room height: floor (H / 0.762 m) units per corner position.

7. Modal Reduction and T60 Prediction

7.1 dB Modal Level Reduction (Heatmap)

The steady-state modal level reduction in dB for one unit group is computed from the effective total absorptance across all placed positions:

$$\Delta L(f) = -10 \times \log_{10}(1 - \alpha\_eff\_total)$$

where α_eff_total is the survival product across all positions:

$$1 - \alpha\_eff = \prod (1 - A\_i) \qquad A\_i \text{ capped at } 0.50$$

Units with α(f) < 0.005 at frequency f are gated out (zero contribution) to prevent off-resonance Lorentzian tails accumulating into spurious broadband reduction figures.

Multiple unit groups (Nodal 30, 55, 70) combine multiplicatively in the power domain:

$$\alpha\_combined = 1 - (1 - \alpha\_40) \times (1 - \alpha\_55) \times (1 - \alpha\_70)$$

$$\Delta L\_combined = -10 \times \log_{10}(1 - \min(\alpha\_combined, 0.97))$$

The 0.97 ceiling reflects a physical constraint: no real panel absorber achieves α = 1.0 at resonance. Edge diffraction losses, frame gaps, mechanical compliance of the membrane perimeter, and material inhomogeneity collectively limit peak in-situ absorption to below 0.97 even under ideal impedance-match conditions (Mechel §11.4; Cox & D'Antonio Ch. 6 measured panel data). This ceiling is separate from, and not redundant with, the saturation cap on δ_TDA, which limits treatment depth relative to the room shell baseline.

7.2 Broadband RT60 — Eyring Equation

Broadband room decay time is computed using the Eyring equation, which corrects the Sabine formula for high absorption:

$$RT60 = 0.161 \times V / (-S \times \ln(1 - \alpha\_total))$$

where V is room volume (m^3), S is total surface area (m^2), and α_total is the area-weighted mean absorption coefficient combining the room shell baseline with all active TDA units:

$$survival = (1 - \alpha\_base) \times \prod\_units\ (1 - \alpha\_i \times A\_unit \times G\_pos / S)$$

$$\alpha\_total = 1 - survival$$

Room shell absorption profiles are based on published ISO 354 reverberation chamber data (Cox & D'Antonio Table 3.3, Everest Table 9.1) with two tiers:

- Sealed / Lab: airtight professional construction; RT60 at 35 Hz typically 2.5–3.5 s
- Residential: elevated effective α at 31.5–50 Hz due to structural leakage paths (HVAC, floor joists, electrical penetrations); RT60 at 35 Hz typically 1.2–1.8 s

7.3 Per-Mode T60 — Modal Loss Budget

Per-mode decay time is computed from the modal loss budget using the same normalization as the heatmap model:

$$T60_n = 13.816 / (\omega_n \times \delta_{total\_n})$$

$$\delta_{total} = \delta_{boundary} + \delta_{TDA}$$

The boundary loss factor uses the total room surface area (Sabine-weighted). From r61, this replaces the earlier per-mode modal coupling area formulation, which used only the two terminating surfaces of each mode. Real room modes lose energy to all six surfaces through structural coupling, wall flexure, HVAC leakage, and transmission loss — not just the two walls the mode is named for. Using ROOM_SA gives physically correct T60 Pre values: residential drywall yields approximately 1.3 s at 30 Hz versus the previous 5.6 s over-prediction. The six-surface orientation weighting (terminating walls 1.0×, parallel surfaces 0.5×) is retained for TDA coupling area calculations in the loss budget — this is distinct from the boundary loss term:

$$\delta_{boundary} = c \times \alpha(f) \times A_{eff}(axis) / (\omega \times 4V)$$

The TDA loss factor per unit group at position ID with stacking count N:

$$\delta_{TDA} \mathrel{+}= (A_{unit} \times G_{pos} \times coupling / A_{eff}) \times stackEffSum(N) \times (\delta_{rad} + \delta_{damp}) \times \alpha_{ratio}$$

where α_ratio = α(f)/α_peak gates off-resonance contribution, and stackEffSum(N) = Σ efficiency(i) for i = 1..N.

This formulation is derived from Kuttruff "Room Acoustics" 5th ed. §3.3 and Morse & Ingard §9.4, and produces T60 values consistent with Eyring when α is small (as expected in the sub-40 Hz range).

7.4 Absorption Saturation Cap

When heavy treatment is applied, δ_TDA >> δ_boundary violates two assumptions of the linear loss-budget model: (1) the undisturbed pressure field assumed by coupling weights, and (2) placement saturation at a finite number of independent modal antinodes.

Saturation is implemented as a ratio cap:

$$\delta_{TDA\_capped} = \min(\delta_{TDA},\ 4.0 \times \delta_{boundary})$$

This caps total decay rate improvement at 5× the bare-room value, giving a minimum T60 floor of approximately T60_bare / 5. For the reference room at 28 Hz (T60_bare ≈ 2.4 s), this yields T60_min ≈ 0.48 s — consistent with the most heavily treated small rooms in published measurement data (Welti 2002, Ryu & Jang 2006, Fazenda & Nicol 2012).

A soft warning threshold is set at δ_TDA > 2.0 × δ_boundary, and the hard cap activates at 4.0×.

7.5 Predicted Measurement Range (Uncertainty)

The linear model overestimates absorption when δ_TDA >> δ_boundary. Uncertainty is one-sided and upward where actual T60 is likely higher than predicted. The width of the predicted measurement range scales with the δ_TDA / δ_boundary ratio:

δ_TDA / δ_boundary	Range Width	Interpretation
0–1×	±5%	Normal model validity, minimal self-loading
1–2×	5→15%	Minor self-loading beginning, slight under-prediction likely
2–4×	15→30%	Meaningful self-loading, validate with REW T60M waterfall
4–6×	30→50%	Substantial uncertainty, model may significantly underestimate T60
>6×	50–70%	Deeply saturated — REW measurement essential

Table 4 — Predicted measurement range vs. saturation ratio. Displayed as T60 [+Xs / ≤Ys] in the simulation.

Note: *These are physics-informed variance estimates, not statistically derived confidence intervals. Use the term 'predicted measurement range' or 'estimated field variance', not 'confidence interval'.*

8. T60 Target Zones

The simulation classifies modal decay into four zones based on published professional monitoring standards and listening room research (EBU R 68, Dolby Atmos room guidelines, Boner & Boner 1966, Toole 2008):

Zone	T60 Range	Description
Over-damped	< 0.25 s	Bass too dry — room sounds unnatural. Typical of studio isolation shells with very heavy treatment.
Target	0.25–0.55 s	Ideal range for professional monitoring. Sufficient decay for bass fundamentals to be audible without smearing.
Moderate	0.55–0.75 s	Audible improvement from treatment — acceptable outcome but not optimal. Common in residential builds with 4–8 units.
Untreated	> 0.75 s	Problematic modal decay. One-note bass, frequency response variance > 15 dB across room positions.

Table 5 — T60 target zones as used in the simulation RT60 canvas

9. PSI AVAA C20 Active Absorber (Optional Module)

The simulation includes a comparative module for the PSI Audio AVAA C20 active low-frequency absorber. The AVAA operates on active pressure zeroing: an internal microphone measures pressure at the unit face, and a driver presents zero acoustic impedance (open boundary condition) at that location.

This mechanism is fundamentally different from the TDA resonant approach:

- Effective range: 15–150 Hz (flat response, no tuning frequency)
- No Q — absorption does not peak/roll off in a Lorentzian sense
- Effective absorption area: 1.4 m² at α = 1.0 (PSI C20 specification, measured at wall boundary)
- Performance rolls off below 25 Hz (speaker excursion limit) and above 120 Hz (spatial aliasing)

The AVAA is modelled using the Sabine loss-factor form — flat across frequency — which correctly represents its broadband character:

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δ_AVAA = A_eff_AVAA × G_pos × coupling × N × eff(f) × c / (ω × 4V ×
A_modal)
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Key distinctions in the model regarding the AVAA:

- AVAA boost factors: G_corner = 2.0, G_wall = 1.0. The PSI specification figure of 1.4 m^2 is measured at a wall boundary and already includes one pressure-doubling (G = 2.0 at a single boundary). Placing the AVAA at a corner adds one further pressure-doubling → effective G_corner = 2.0 relative to the specification figure, equivalent to G = 4.0 relative to free-field — the same total gain as for a passive TDA. The apparent asymmetry between the TDA G_corner = 4.0 and AVAA G_corner = 2.0 is purely a reference-frame difference: TDA boost applies to raw panel area (no pre-baked boundary gain); AVAA boost applies to the PSI specification area which already includes one wall doubling.
- AVAA δ_AVAA is NOT subject to the TDA saturation cap (active mechanism, not passive Sabine saturation)
- A harder uncertainty cap of 1.5 × δ_tda ratio weight is applied to AVAA contributions in the combined confidence interval, reflecting that the AVAA model does not account for multi-unit coupling, placement variation, or room-specific boundary conditions — sources of uncertainty that are larger than those in the TDA model at the same treatment ratio.
- For practical decisions: the model correctly indicates that one AVAA in a corner will meaningfully reduce modal decay in the 25–100 Hz range, and that two units will reduce it further. Beyond that, treat the exact T60 numbers as directional.

10. Schroeder Frequency and Model Applicability

The Schroeder frequency marks the transition between the modal regime (discrete, identifiable resonances below) and the statistical regime (diffuse field, overlapping modes above):

$$f\_s \approx 2000 \times \sqrt{(RT60 / V)}$$

For the reference room with RT60 ≈ 1.5 s and V ≈ 71.6 m³, f_s ≈ 80 Hz. The TDA model is most accurate in the modal regime below this frequency. Above f_s, treatment predictions become less meaningful as modes overlap and the diffuse-field model takes over.

10.2 Sub-Fundamental Frequency Floor

A lower bound on model validity is equally important. The loss-budget model requires the existence of a standing wave at frequency f. Below the room's lowest axial mode — the sub-fundamental floor f_1_min — no such standing wave exists:

$$f_1\_min = 1.1 \times c / (2 \times \max(L, W, H))$$

The factor 1.1 provides a 10% guard band above the strict theoretical cutoff. Below f_1_min the room acts as a pressure vessel and decay is governed by structural leakage and radiation losses, not boundary absorption. Predictions in this regime are physically meaningless and should not be used for design decisions.

For the reference room (L = 6.1 m), f_1_min ≈ 31 Hz. The model displays a warning and shaded region in both the dB vs Frequency and Bass Modal Decay canvases when the selected frequency falls below this floor. The MODES table is restricted to frequencies at or above the first computed axial mode.

Note: *Predictions below f_1_min may appear numerically plausible but have no physical basis. The model extrapolates from boundary absorption alone in a regime where the diffuse-field assumption is doubly invalid.*

The simulation displays the Schroeder frequency as a guide but does not enforce a hard cutoff — the transition is gradual and room-dependent.

11. Validation, Calibration, and Limitations

11.1 Calibration Targets

Model parameters are calibrated against published data and typical REW measurements:

- Plasterboard drywall at 60–80 Hz: mean α ≈ 0.10–0.12 (Cox & D'Antonio Table 3.3, Everest Table 9.1)
- Untreated 16×12×8 ft room at 60 Hz: RT60 ≈ 0.86 s; 20×16×8 ft: RT60 ≈ 0.98 s
- 4-unit corner treatment: RT60 reduction ~30–35% vs. untreated
- 8-unit corner treatment: RT60 reduction ~45–50% vs. untreated
- Most heavily treated published rooms: T60_min ≈ T60_bare / 5 (Welti 2002, Fazenda & Nicol 2012)

11.2 Known Limitations

The model makes the following simplifying assumptions that users should be aware of:

- Rectangular room geometry only; no correction for non-rectangular plans or structural irregularities
- Axial modes modelled explicitly; oblique and tangential modes treated as background (they are typically lower Q and less problematic than axial modes)
- Pressure field undisturbed by treatment — valid at low unit counts but breaks down at saturation (δ_TDA/δ_boundary > 4)
- No inter-unit acoustic coupling beyond stacking efficiency factors, multiple units at different positions are combined assuming statistical independence
- Construction profiles use area-weighted mean α across all six surfaces, frequency-specific per-surface variation is not modelled
- Residential leakage paths (HVAC, floor joists, electrical) appear as elevated effective α, not as true acoustic absorption as the model cannot separate these
- Predictions below the sub-fundamental floor (f_1_min ≈ 31 Hz for reference room) are outside model validity — the loss-budget model requires an existing standing wave to couple to. A warning is displayed but predictions are still computed for reference.
- Q < 0.8 bandwidth correction (k_bw = 1.8) is an empirical calibration parameter without a direct published source. Treat T60 and α predictions for over-damped units (Q < 0.8) as provisional pending in situ REW measurements.
- Stacking efficiency factors [1.00, 0.90, 0.82, 0.75] are calibrated from high-frequency panel data (Fuchs & Zha, 200–500 Hz) and are not validated for LF TDA arrays below 100 Hz. Near-field coupling at sub-100 Hz wavelengths may be stronger than these values suggest.

11.3 Axial-Only Modelling — Scope and Implications

The model explicitly predicts T60, dB reduction, and heatmap pressure distribution for axial modes only — standing waves along a single room dimension. This is a deliberate scope choice, not an oversight, and its accuracy implications depend strongly on frequency range.

11.3.1 Mode Population in the Reference Room (20–150 Hz)

In the reference room (6.1 × 4.9 × 2.4 m), the approximate mode count by family across 20–150 Hz is:

Mode Family	Count (approx.)	Model Status
Axial (one axis)	12–15	Fully modelled — T60, dB, heatmap
Tangential (two axes)	30–45	Informational display — frequency only, no T60
Oblique (three axes)	20–35	Informational display — frequency only, no T60

Table 6 — Mode family count and model coverage for the reference room, 20–150 Hz

11.3.2 Why Axial Modes Dominate Audible Problems

Although axial modes represent only 15–25% of the total mode count, they are responsible for most audible bass problems in small rooms for three reasons:

- Highest Q: axial modes reflect from only two opposing surfaces per cycle, accumulating less boundary loss per round trip than tangential (four surfaces) or oblique (six surfaces) modes. Natural T60 at resonance is therefore longest for axial modes.
- Largest pressure variation: axial mode pressure swings from P_max at both terminating walls to zero at the mid-plane, creating the 20 dB+ seat-to-seat SPL variation characteristic of untreated small rooms. Tangential and oblique pressure distributions are more spatially diffuse.
- Lowest frequency onset: the first axial modes occur at c/(2L), c/(2W), and c/(2H). The first tangential mode in the reference room is ~45 Hz; the first oblique is ~63 Hz. Below 45 Hz the modal landscape is almost exclusively axial — precisely the range where the model is most used and most critical.

Note: *Published research (Toole 2008, Walker 1996, Welti 2002) consistently identifies axial modes as the primary source of perceptual problems in rooms below 80 Hz. Tangential and oblique modes are typically 3–6 dB lower in steady-state boundary amplitude due to reduced pressure doubling.*

11.3.3 Error Direction

Omitting tangential and oblique modes from the treatment model introduces errors in specific directions:

- T60 predictions (per-mode): conservative. Corner-placed TDA units absorb tangential modal energy that the model does not credit. Measured post-treatment T60 at a given frequency will reflect decay from all mode families, not just the axial mode. For frequencies where axial and tangential modes overlap, the model may overpredict T60, and real treatment is more effective than predicted.
- Heatmap pressure display: axial-only. The spatial pressure pattern is computed solely from axial modes. Above ~60 Hz, where tangential modes multiply rapidly, the displayed pattern may not reflect the actual sound field. A position shown as a pressure node for an axial mode may simultaneously be a pressure maximum for a dominant tangential mode.
- Mode frequency landscape: previously underreported. Without tangential and oblique frequencies visible, a user observing a measured RT60 peak at a frequency with no listed axial mode might incorrectly conclude there is no modal explanation. From r50 onward, all three mode families are shown in the mode table as informational rows.

11.3.4 Accuracy by Frequency Range

Range	Dominant Family	Model Accuracy Implication
20–45 Hz	Axial only	High — modal landscape is almost exclusively axial. Per-mode T60 and heatmap are reliable guides.
45–80 Hz	Axial + tangential	Moderate — tangential modes present but axial modes are typically dominant. T60 may be slightly overestimated (conservative error). Validate with REW.
80–150 Hz	All families	Lower — above the Schroeder frequency, overlapping modes make axial-only T60 a poor proxy for measured decay. Use the broadband Eyring RT60 in preference to per-mode values.

Table 7 — Model accuracy by frequency range relative to modal family dominance

11.3.5 Mode Display Convention (r50+)

From r50, the mode table in the simulation UI displays all three families across 20–150 Hz in a single merged, frequency-sorted list:

- Axial modes: full color, clickable, shows T60 before/after treatment and dB reduction estimate
- Tangential modes: greyed italic rows with tooltip — informational only, no T60 or dB data. Corner TDA placement provides partial coupling to these modes even though the model does not quantify it.
- Oblique modes: further greyed italic rows — informational only. Typically lowest Q and best naturally damped of the three families.
- Mode count footer: e.g. '13 axial (modelled) · 38 tangential · 27 oblique (informational) · 20–150 Hz'

This allows users to identify when a measured RT60 peak at a given frequency is likely driven by a tangential or oblique mode — providing an explanation even where the model cannot quantify the treatment effect.

11.4 Recommended Validation Protocol

The simulation tool is intended for design guidance, not engineering certification. All predictions should be validated with in-situ measurements. From r61, the REW waterfall T60M import (Section 12) computes T60M at all mode frequencies directly from waterfall exports. From r58, the Treatment Progress panel provides per-mode target status and unit-count projections; listening position analysis shows modal pressure and T60 at the mix seat; and configurations can be saved and shared via URL hash.

11.4.1 Measurement Setup

- Use REW (Room EQ Wizard) with a calibrated measurement microphone. A Class 1 or Class 2 SPL meter microphone is sufficient; a calibrated measurement microphone (e.g. miniDSP UMIK-1) is preferred.
- Use a swept-sine stimulus (not MLS) for all bass frequency measurements. Swept-sine provides 20–30 dB better SNR at sub-50 Hz frequencies and is essential for clean T60 data below 40 Hz.
- Set REW Analysis Preferences → Octave Smoothing to None and resolution to 1/48-octave or 1/24-octave. This is required for waterfall T60M resolution adequate to isolate individual axial modes.
- Ensure at least 35–40 dB of dynamic range in the decay window for each frequency of interest. REW flags noise-floor limited measurements — do not import T60 values from noise-floor limited bands.

- Measure at the primary listening position and at least two additional positions near room boundaries to capture spatial variation.

11.4.2 REW Export for Import

- In REW: Waterfall graph → configure Mode=Fourier, Window=300ms, Rise Time=30ms, Total Slices=1000, Smoothing=1/48 oct, Normalize unchecked, CSD unchecked → File → Export → Export data as text (.txt). Both measurements must use identical settings.
- File → Export → Export RT Parameters as text. Save the .txt file. This is the file accepted by the model's REW import panel.
- For the untreated baseline: export before any TDA units are installed. For the treated validation: export after all intended units are installed and settled (allow 24 hours in heated rooms for panel compliance to stabilize).

Note: *The import parser accepts tab-delimited and comma-separated REW exports across multiple REW version formats.*

11.4.3 Calibration and Validation Workflow

- Step 1 — Configure construction profile: select the correct construction profile (res_drywall or res_treated) and set coverage % and panel depth if applicable. These parameters affect T60 Pre and the broadband panel contribution table in the PDF report. T60M is computed automatically at all axial mode frequencies. The model auto-solves G_corner by minimizing mean-squared log-error between predicted and measured T60M at all axial mode frequencies. Review the per-mode residual table; MAE < 0.15 s indicates a reliable calibration.
- Step 2 — Import untreated waterfall: export the REW waterfall before treatment and load into the Untreated slot. T60M is computed at all axial mode frequencies. Step 3 — Apply G_corner: click Apply G = X. The slider, badge, and all predictions update immediately. The Bass Modal Decay canvas now shows the continuous measured T60M curve (green) and square markers at each axial mode frequency, with model diamonds for comparison.
- Step 3 — Design treatment: with G_corner calibrated, add TDA units (Nodal 30 and Custom Unit 2). All T60 predictions are now anchored to the measured room.
- Step 4 — Install units and re-measure: after installation, export the treated REW T60M and load it into the Treated slot. The canvas adds a second measured curve (lime) with circles at axial mode frequencies. The mode table gains a measured column showing interpolated T60 at each axial mode with a percentage residual vs. the model's treated prediction.
- Step 5 — Iterate: where residuals are large (> 20%), investigate: check whether the frequency is near a tangential mode (visible in the mode table), whether the saturation warning is active, or whether the construction profile needs adjustment.

- Step 6 — Validate SPL distribution: record heatmap position measurements to validate spatial predictions. The heatmap pressure display is axial-only above 60 Hz, tangential modes will cause measured SPL patterns to diverge from the display.

12. REW Waterfall T60M Import (r61+)

From r61, the simulation includes a REW waterfall T60M import panel that transforms the model from a standalone design aid into a calibrated validation instrument. The feature accepts REW waterfall exports (.txt) and computes T60M at every axial room mode frequency using a Nelder-Mead STFT decay fitting engine — replacing the previous RT Parameters T30 import. It serves four purposes: G_corner auto-calibration from an untreated measurement, visual overlay of the continuous measured T60M curve across 20–150 Hz in the Bass Modal Decay canvas, square markers at each axial mode frequency, and a measured column in the per-mode T60 table.

12.1 Why Waterfall T60M

The model's per-mode architecture operates at exact axial mode frequencies — for example, X1 at 28.1 Hz, Y1 at 35.0 Hz. Meaningful comparison between model predictions and measurements requires frequency resolution fine enough to resolve these as separate peaks.

The REW waterfall export provides full-spectrum decay data at every frequency bin (typically 1067 bins from 10–22 kHz). The T60M engine fits a Nelder-Mead model to the decay at each frequency, using a configurable Rise Time (default 30 ms, matching REW setting). This gives T60M at every room mode frequency without requiring manual frequency alignment or narrow-band export configuration.

The waterfall method computes T60M from the actual decay shape at each frequency using Nelder-Mead curve fitting, making it independent of the T20 vs T30 trade-off. REW Waterfall settings: Mode=Fourier, Window=300ms, Rise Time=30ms, Total Slices=1000, Smoothing=1/48 oct, Normalize unchecked, CSD unchecked. The Rise Time setting must match the value entered in the model import panel.

Note: *REW flags noise-floor limited measurements. Do not import T60 values from noise-floor limited frequency bins — they will appear artificially short and produce an over-optimistic G_corner calibration.*

12.2 G_corner Auto-Calibration Solver

G_corner (default 4.0, range 1.0–8.0) is the single most influential model parameter — it scales every T60 and dB prediction. The import solver finds the G_corner that minimizes the residual between predicted and measured T60M across all axial mode frequencies where measurement data exists within ±4 Hz.

The objective function is mean-squared log-error:

$$E(G) = (1/N) \times \Sigma_i \; [\ln(\text{T60\_model}(G, f_i)) - \ln(\text{T60M\_meas},i)]^2$$

Log-error is used rather than absolute error because it treats a factor-of-two discrepancy the same way regardless of whether T60 is 0.5 s or 2.0 s. This is appropriate for a loss-budget model whose predictions scale proportionally with G.

The solver uses golden-section search over $G \in [1.0, 8.0]$ with 60 iterations, converging to within 0.001 of the true minimum. No gradient is required. The optimum is unique because T60_model increases monotonically with G (more pressure gain → more absorption → shorter T60) while the measurement is fixed.

After solving, the panel reports:

- Suggested G value (rounded to one decimal place)
- Number of axial modes used in the fit
- Mean absolute error in seconds with quality rating: Good (MAE < 0.15 s), Moderate (MAE < 0.35 s), Poor (MAE ≥ 0.35 s, check room geometry or construction profile)
- Per-mode residual table showing measured, predicted, and percentage error for up to six axial modes

Note: *A Poor fit (MAE ≥ 0.35 s) most commonly indicates that the room construction profile is mismatched (e.g. selecting Sealed when the room has significant residential leakage), or that the room geometry is non-rectangular enough to shift actual mode frequencies away from the rectangular room prediction. Adjust the construction profile first, then re-import.*

12.3 Parser

The parser accepts REW waterfall text exports in comma-separated semicolon/space. Column detection is name-driven as it locates Frequency and SPL.

Metadata extraction: measurement label and date are read from the comment block where present and displayed in the status area. Data is filtered to 20–200 Hz.

Frequency interpolation (rewInterp): given a query frequency, the function linearly interpolates between the two nearest bracketing data points. It returns null if the query is more than 3 Hz outside the measured frequency range, preventing extrapolation beyond available data.

12.4 Bass Modal Decay Canvas — Overlay Display

When measurement data is loaded, the Bass Modal Decay canvas adds up to two additional curves alongside the existing model predictions:

Curve	Color	Description
T60 Untreated (model)	Orange	Per-mode modal loss-budget T60 at each axial mode; open orange diamonds
RT_{60} Treated (model)	Cyan	Per-mode modal loss-budget T60 with TDA placement; zone-coloured circles
REW T60 Untreated (measured)	Yellow-green	Continuous T60M curve 20–150 Hz (green); filled squares at each mode frequency
REW T60 Treated (measured)	Lime	Continuous T60M curve 20–150 Hz (blue); filled squares at each mode frequency; zone-coloured model circles overlaid for live comparison
◇ Per-mode untreated (model)	Orange outline	Modal loss-budget T60 at each axial mode frequency, no treatment
● Per-mode treated (model)	Colored by zone	Modal loss-budget T60 at each axial mode frequency with TDA treatment

Table 8 — Bass Modal Decay canvas curves when REW waterfall T60M data is imported

12.5 Mode Table — Measured Column

Each axial mode row in the mode table gains a sub-row when REW data is loaded. It shows:

- M: prefix (yellow-green) followed by the measured T60M at the mode frequency, color-coded by the standard zone thresholds (< 0.25 s red, ≤ 0.55 s green, ≤ 0.75 s amber, > 0.75 s orange)
- Percentage residual between the measured T60M and the closest model prediction (pre-treatment T60 if only untreated data is loaded; post-treatment T60 if treated data is also loaded). Residual shown in green if < 15%, amber if ≥ 15%.

This allows immediate identification of modes where the model diverges significantly from measurement. A large positive residual (model predicts shorter T60 than measured) indicates over-prediction of treatment effect, which is common near the saturation boundary or where stacking efficiency is overestimated. A large negative residual (model predicts longer T60 than measured) may indicate a tangential or oblique mode contributing to measured decay at that frequency.

12.6 Interpretation Guidance

The following patterns in the overlay display indicate specific model or measurement conditions:

- Model and measured untreated T60M agree closely (< 10%) across 28–70 Hz: G_corner is well calibrated and the construction profile is appropriate. Proceed to treatment design with confidence.
- Model untreated T60 consistently lower than measured T60M across all frequencies: G_corner too low — the model underestimates room pressure gain. Apply the solver's suggested G or increase G_corner manually.
- Model untreated T60 consistently higher than measured T60M: G_corner too high, or the room has more structural leakage than the selected construction profile implies. Try switching to a residential construction profile or reduce G_corner.
- Good agreement at low frequencies (28–45 Hz) but divergence above 60 Hz: tangential mode contribution increasing above 60 Hz which is expected behavior. The axial-only model is valid in the lower range; use the measured data to calibrate treatment targets above 60 Hz directly.
- Post-treatment measured T60M substantially above model treated prediction: saturation cap is binding, or stacking efficiency is overestimated. The measured data is the ground truth, do not add further treatment units without re-measuring.
- Post-treatment measured T60M below the model prediction at a specific mode: the model is conservative (as expected for corner-placed TDAs that also absorb tangential energy). This is the correct error direction where the actual treatment is more effective than predicted.

13. Non-Rectangular Room Geometry: Sloped and Cathedral Ceilings

From r55, the simulation supports non-rectangular vertical geometry through a first-order perturbation model. Two ceiling profiles are implemented: a single-ramp sloped ceiling (one inclined plane rising from H_1 at one end to H_2 at the other) and a symmetric cathedral ceiling (two inclined planes rising from eave height H_1 at both side walls to ridge height H_2 at the center). The rectangular room model is a limiting case of both profiles when $H_1 = H_2$, ensuring full backward compatibility.

13.1 Perturbation Theory Basis

The rectangular room modal model relies on closed-form eigenfunctions — $\cos(n\pi x/L)$ — which have no analogue for arbitrary geometry. A general solution requires numerical methods (FEM or FDM), which are computationally impractical for browser-interactive use at these frequencies. Perturbation theory provides a tractable analytical correction by treating the non-rectangular room as a rectangular room (with height $H\_avg = (H_1 + H_2) / 2$) subject to a small geometric perturbation.

The perturbation correction is grounded in the modal loading integral from Morse & Ingard ("Theoretical Acoustics" §9.4) and Kuttruff ("Room Acoustics" 5th ed., §2.4). For a room with a linearly varying ceiling height, the first-order frequency correction to Z-axis mode n is:

$$\mathbf{f\_corrected = f\_rect \times \sqrt{(1 - \varepsilon)} \quad where\ \varepsilon = (1/12) \times (\Delta H / H\_avg)^2}$$

where $\Delta H = |H_2 - H_1|$ is the total ceiling height variation and H_avg is the arithmetic mean height. f_rect is the frequency of the corresponding mode in a rectangular room of height H_avg. The correction is negative — a sloped ceiling lowers Z-axis mode frequencies slightly relative to the naive H_avg substitution. The factor (1/12) arises from the integral of the squared cosine mode shape over the linearly varying boundary displacement.

For the symmetric cathedral profile, the perturbation integral over the piecewise-linear ceiling yields the same ε formula applied to the half-span rise $(H_2 - H_1)$ over half the room span. The two halves of the ceiling contribute coherently to the first-order correction, giving the same expression. The correction is applied to all Z-axis modes n = 1, 2, 3, … with a minimum floor of $\sqrt{0.5} \approx 0.707$ on the correction factor to prevent non-physical results at extreme slope angles.

X-axis and Y-axis mode frequencies are not corrected. The slope coupling correction for horizontal modes is proportional to $(\Delta H/2H\_avg)^2$, which is below 0.1% for all slope angles under 20° — negligible for treatment planning. Only Z-axis modes are affected by a sloped ceiling at practical room proportions.

13.2 Room Geometry: Volume, Surface Area, and Effective Height

For a linearly sloped ceiling, the room volume is exact at V = L × W × H_avg. This follows directly from the prismatic cross-section: the sloped ceiling traces a linear gradient, and the average height across the floor plan is the arithmetic mean. For the symmetric cathedral profile the same formula applies: V = L × W × H_avg, where H_avg = $(H_1 + H_2) / 2$. This is exact for a symmetric tent.

Surface area is computed from the actual geometry of each surface, not from the flat-ceiling approximation. For the single-ramp profile with slope running along the length axis:

$$A\_ceil = W \times \sqrt{L^2 + \Delta H^2} \quad \text{(sloped ceiling panel area)}$$

For the cathedral profile, two symmetric panels of half-span replace the single panel:

$$A\_ceil = 2 \times W \times \sqrt{(L/2)^2 + \Delta H^2} \quad \text{(two symmetric roof panels)}$$

Gable end walls (cathedral, perpendicular to the ridge axis) are trapezoids of average height H_avg. Side walls (parallel to the ridge) are rectangles of the eave height H_1, since the slope is internal to the roof structure. The floor remains L × W. These areas feed into both the Eyring RT60 calculation (Section 7.2) and the six-surface modal coupling area (Section 6.3). Using the exact sloped surface area rather than the flat-ceiling approximation increases S by 0.5–3% for typical studio room slopes, modestly reducing predicted RT60 and increasing boundary loss δ_boundary.

13.3 Spatial Pressure Function and Heatmap

In the rectangular model, Z-axis mode n has the spatial pressure function cos(nπz/H), evaluated at a fixed room height H. For a sloped or cathedral ceiling, the local ceiling height varies with floor-plan position. The perturbation model replaces the fixed height with the local height H(x, y) at each point:

$$P_2(x, y, z) = \cos(n\pi z / H(x, y))$$

Single ramp: $H(x, y) = H_1 + (H_2 - H_1) \times x/L$ (slope along length axis)

Cathedral: $H(x, y) = H_1 + (H_2 - H_1) \times (1 - |2x/L - 1|)$ (symmetric tent, ridge at x = L/2)

This function correctly moves the pressure nulls and antinodes of Z modes with the ceiling geometry across the floor plan. For the cathedral profile the Z1 pressure maximum appears at the floor (z = 0) and at the ridge ($x = L/2$, $z = H_2$) simultaneously, and decreases symmetrically toward both eave walls — reflecting the physical standing wave in the tent-shaped space. For the single-ramp profile the effective cavity depth increases monotonically from the low end to the high end, shifting the pressure distribution accordingly.

In the heatmap cross-section views (XZ and YZ planes), pixels above the local ceiling height H(x, y) are masked and rendered as the background colour. The ceiling boundary is drawn as a dashed orange line indicating the sloped or cathedral profile. X-axis and Y-axis mode spatial functions are unchanged — cos(nπx/L) and cos(nπy/W) respectively — because horizontal mode shapes are not materially affected by ceiling slope at practical room proportions.

13.4 Modal Coupling Area and TDA Position Weighting

The six-surface modal coupling area for Z-axis modes (Section 6.3) uses the actual sloped or cathedral ceiling area rather than the flat-ceiling area L × W. The terminating surfaces for a Z-axis mode are the floor and the ceiling. For a sloped ceiling these are no longer the same area. The floor contributes L × W; the ceiling contributes its actual inclined surface area as computed in Section 13.2. The parallel surface contribution (A_para = ROOM_SA − A_term) is updated accordingly. This increases the effective modal coupling area relative to a flat ceiling of the same H_avg, which slightly reduces predicted boundary loss δ_boundary and increases predicted untreated T60 — a correct physical result, since the sloped ceiling is a less efficient sound reflector per unit floor area than a flat ceiling.

TDA position coupling weights for Z-axis modes are adjusted to reflect the asymmetry of the sloped ceiling geometry. In the flat-ceiling model, all four floor corners are equivalent for Z-mode coupling. In the sloped-ceiling model:

- Single ramp: the local ceiling height H(x, y) at each corner is used to compute a coupling weight proportional to H_local / H_avg. Corners at the tall-wall end have higher local ceiling height and therefore higher Z-mode coupling; corners at the low end have reduced coupling. The weight is bounded between 0.5 and 1.0.
- Cathedral: all four floor corners are at eave height H_1 equally. The Z-mode coupling weight is uniform across all corners, computed as 0.75 × (H_1 / H_avg), bounded between 0.5 and 0.75. Eave corners couple less strongly to Z modes than in an equivalent flat-ceiling room because the pressure antinodes are shared between the floor and the ridge — a location with no TDA placement — rather than between the floor and a flat ceiling of uniform height.

X-axis and Y-axis mode coupling weights are unaffected by ceiling slope in either profile, because their terminating surfaces are the end walls — which remain vertical rectangles — and their pressure distribution is determined by the horizontal room dimensions, not the ceiling height.

13.5 Validity Limits

Perturbation theory is valid when the geometric deviation is small relative to the room dimension: $\Delta H / H\_avg \ll 1$. The correction $\varepsilon = (1/12)(\Delta H/H\_avg)^2$ is quadratic in this ratio, so accuracy degrades rapidly at large slopes. The model classifies slope angle and warns accordingly:

Panel slope angle	Status	Implication
≤ 15°	Valid	Perturbation accurate to ~5% on Z-axis mode frequencies. Mode frequency corrections are small (typically < 1 Hz). Treatment planning results reliable.
15° – 25°	Caution	Perturbation results are directional, not quantitative. Second-order terms become non-negligible. Mode frequencies may differ from predicted by 2–5 Hz. Validate with REW measurement.
> 25°	Invalid	Perturbation approximation degrades significantly. Use H_avg as a flat-ceiling approximation for treatment planning. The tool displays a warning and the model reverts to H_avg behaviour.

Table 9 — Sloped/cathedral ceiling perturbation validity by panel slope angle. Angle is measured from horizontal to the ceiling panel surface; for cathedral ceilings, the angle is measured on each half-panel (eave to ridge over half the room span).

The dominant practical use case — residential rooms with 6:12 to 8:12 roof pitch (26°–34°) converted to studio use — falls at the boundary of the caution and invalid zones. For these rooms, H_avg should be used as a rectangular approximation for treatment planning, and the REW G_corner calibration workflow (Section 12) should be employed to anchor model predictions to measured T60M before committing to a unit configuration. The perturbation model is most reliable for shallow commercial slopes (3:12 to 5:12, or 14°–23°) and purpose-built studio rooms with modest ceiling rake.

13.6 Treatment Planning in Sloped and Cathedral Rooms

Several practical differences apply when designing treatment for rooms with non-rectangular vertical geometry.

Z-axis mode frequencies shift with H_{avg}. The first Z-axis mode is $f_{Z1} = c / (2 \times H_{avg})$. In a room with 2.4 m eave and 3.6 m ridge, $H_{avg} = 3.0$ m and $f_{Z1} = 57.2$ Hz — compared to 71.5 Hz for the same room with a flat 2.4 m ceiling. This lower f_{Z1} moves the height mode into the primary treatment range of the Nodal units, making it more important to identify and address. The corrected Z-axis mode frequencies are displayed in the mode list annotated with a "~sloped" or "~cathedral" tag and a tooltip showing the rectangular reference frequency.

Corner placement priority changes for single-ramp ceilings. In a flat-ceiling room, all four floor corners are equivalent for Z-mode coupling. In a sloped room, the two corners at the tall-wall end provide higher Z-mode coupling (coupling weight up to 1.0) while the corners at the low-wall end provide reduced coupling (down to 0.5). For rooms with a dominant Z-mode problem, units at the tall-end corners are more effective per unit. The simulation displays differentiated T60 predictions for each corner position when the slope is enabled, enabling informed corner selection.

For cathedral ceilings, the ridge creates a pressure maximum for Z modes at the center of the room in addition to the floor. TDA units cannot be placed at the ridge, so the treatment is restricted to floor and eave-wall positions. The model accounts for this by computing coupling only from accessible floor-corner positions. The treatment efficiency for Z modes in a cathedral room is therefore lower per unit than in an equivalent flat-ceiling room, and a larger number of units may be required to achieve the same T60 reduction target.

As with all simulation results, predictions for sloped and cathedral rooms should be validated against REW T60M measurements using the G_{corner} calibration workflow described in Section 12. The perturbation model introduces additional approximation beyond the axial-mode simplifications documented in Section 11. The REW calibration step is particularly important in these rooms, because the G_{corner} that minimizes residual error may differ from the standard 4.0 default as a result of the modified pressure field geometry.

14. Simulation Tool — r61 Features

14.1 Treatment Progress Panel

A Treatment Progress panel replaces the per-position T60 breakdown in r58. The panel displays a per-mode table showing pre-treatment and post-treatment T60, color-coded status against a user-configurable target (default 0.55 s), the best-matched unit type for each mode (highest α at that mode frequency), and forward projections of T60 if additional units of the best type were added. Projections distribute hypothetical units across corners in round-robin order and run the full modal T60 calculation on the resulting configuration. The saturation indicator uses the model's existing δ_tda cap: when δ_tda ≥ SATURATION_RATIO_MAX × δ_boundary (atFloor = true), the panel reports that additional units cannot further reduce T60 and displays the T60 floor value. This is a stable, physics-based condition rather than a marginal-gain threshold.

14.1 Listening Position Analysis

Clicking or dragging on the floor plan heatmap places a listening position marker. The marker displays room coordinates (x, y in meters) on the canvas. A per-mode T60 table in the sidebar panel shows, for each axial mode: frequency, post-treatment T60 (color-coded against the target), AT POS percentage showing the fraction of peak pressure at that location (computed as |cos(nπ·pos/dim)| for the mode's axis), and a HEARD label (STRONG / PARTIAL / WEAK) indicating practical audibility at the listening seat.

WEAK (AT POS < 35%) indicates the listening position is near a pressure null for that mode. This reduces steady-state excitation at the seat during sustained notes but does not eliminate perception of the modal decay tail after note cessation. Modal decay is a diffuse-field phenomenon: the decaying energy releases from wall boundaries and fills the room regardless of the listener's position relative to the standing wave's spatial pattern. A mode with long T60 and WEAK pressure at the listening position will still produce an audible decay tail on transients. Treatment of that mode remains warranted for mixing accuracy on bass transients even when the steady-state pressure at the seat is low.

14.3 AVAA Confidence Band Revision

In r57 and earlier, the combined uncertainty fraction for hybrid TDA+AVAA configurations weighted the AVAA ratio at 0.5× TDA, implying AVAA predictions were better characterized than TDA predictions. This was reversed in r58. The AVAA single-unit model uses PSI's published effective control area and does not account for multi-unit coupling, placement variation, or room-specific boundary reflections. The AVAA contribution to uncertainty is now weighted at 1.5× TDA ratio, producing a wider confidence interval for hybrid configurations. Combined TDA+AVAA predictions should be treated as more uncertain than TDA-only predictions at the same aggregate treatment ratio.

14.5 Configuration Save/Load Local and via URL Hash

From r58, the “Share / Save Config” button encodes the full simulation state including room dimensions, construction profile, all unit tunings and positions, G_corner, cavity depth, membrane dimensions, slope settings, and view state — as base64 JSON in the browser URL hash. Sharing a configuration requires only copying the URL from the address bar. Loading a shared configuration requires navigating to that URL; the tool restores the full state on page load. Sliders auto-save to the hash with a short debounce after any change. This replaces the JSON export workflow for configuration sharing while retaining the JSON export for archival purposes.

Saving — clicking "+ Save current" prompts for a name, pre-filled with a suggested name derived from the current room dimensions and unit configuration (e.g. "6.1×4.9×2.4m 4×N40 4×N50"). Each saved entry stores the full state hash, the room description, unit description, and timestamp. If you save with a name that already exists it overwrites that entry.

Loading — clicking a configuration name restores the full state exactly including room dimensions, construction, all unit tunings and positions, G_corner, cavity depth, membrane dimensions, slope settings, view state, and re-renders everything.

Renaming — a "rename" button on each row prompts for a new name with duplicate detection.

Deleting — a red ╳ button with a confirmation prompt removes the entry.

Metadata — each row shows the room dimensions, unit summary, and a human-readable age ("today", "yesterday", "3d ago", "2mo ago") so you can tell at a glance how old each configuration is.

Empty state — when no configurations are saved the panel shows "No saved configurations yet. Click '+ Save current' to save one."

14.6 Part-Treated Construction Profiles — Coverage and Depth

14.6.1 Background and Motivation

Prior to r61, the two part-treated construction profiles stored fixed alpha arrays computed from a single assumed configuration: 30% surface coverage of 4″ OC 703 panels. While the area-weighted mixing formula was correct, the baked-in values prevented the model from reflecting the wide variation in real studio installations (panel coverage ranging from

10% to 50%), and panel thickness spanning 2″ through 6″. Both parameters have a substantial effect on the computed T60 Pre, and therefore on the predicted T60 Post improvement attributable to TDA treatment.

14.6.2 Alpha Computation Model

From r61, when a part-treated profile is selected, baselineAlpha(f) computes effective alpha dynamically rather than reading from a fixed array:

```
α_eff(f) = (1 − coverage) × α_shell(f) + coverage × α_OC703(f,depth)
```

where coverage is the fraction of total room surface area covered by panels (0.05–0.60, default 0.30), depth is the panel thickness in inches (2, 4, or 6), α_shell(f) is the bare room shell absorption interpolated log-linearly from the construction profile, and α_OC703(f, depth) is the Owens Corning OC 703 published absorption coefficient at that octave band and thickness from ISO 354 data. Interpolation between octave bands uses log-linear frequency scaling consistent with the existing profile interpolation function (interpAlphaLog).

The OC 703 absorption data embedded in the model (OC703_BY_DEPTH) is as follows:

14.6.3 OC 703 Absorption Data by Thickness (ISO 354)

Table 14.1 — Owens Corning OC 703 absorption coefficients by octave band and thickness. Values are ISO 354 published data used directly in the model without scaling.

Depth	31.5 Hz	50 Hz	63 Hz	80 Hz	125 Hz	Practical lower limit
2″	0.04	0.06	0.08	0.11	0.28	~125 Hz
4″ (default)	0.09	0.25	0.43	0.62	0.97	~50 Hz
6″	0.18	0.45	0.77	0.95	1.00	~31.5 Hz

"Practical lower limit" = lowest octave band where α ≥ 0.15. Below this frequency, panel contribution is negligible relative to room shell losses and TDA absorption dominates.

14.6.4 Quantitative Effect of Coverage and Depth on T60 Pre

Table 14.2 shows α_eff and T60 Pre at key octave bands for a reference room (6.1 × 4.9 × 2.4 m, V = 71.5 m³, SA = 122.8 m²) with res_drywall shell construction and 30% surface coverage. The computed T60 uses the Sabine-weighted boundary damping formula:

```
δ_boundary = c × α_eff × ROOM_SA / (ω × 4 × V)
```

Freq	α shell	T60 shell	T60 2″ 30%	T60 4″ 30%	T60 6″ 30%	T60 6″ 50%	Target
31.5 Hz	0.070	1.467 s	1.467 s	1.351 s	0.997 s	0.821 s	
50 Hz	0.090	1.141 s	1.141 s	0.744 s	0.518 s	0.380 s	
63 Hz	0.130	0.790 s	0.790 s	0.467 s	0.319 s	0.228 s	✓
80 Hz	0.140	0.733 s	0.733 s	0.361 s	0.268 s	0.188 s	✓
125 Hz	0.068	1.510 s	0.780 s	0.303 s	0.295 s	0.192 s	✓

Table 14.2 — T60 Pre at key octave bands for res_drywall shell + 30% panel coverage (and 50% for 6"). Green values ≤ 0.55 s (target range). Orange values > 1.0 s. Target column indicates whether 4" panels at 30% alone achieve the target without TDA treatment. Reference room: 6.1 × 4.9 × 2.4 m.

14.6.5 Design Implications for TDA Requirement

The data in Table 14.2 make the complementary roles of panels and TDAs explicit. At 31.5 Hz, even 6″ panels at 30% coverage produce T60 = 0.997 s — well above target. Increasing coverage to 50% reduces this to 0.818 s, still above target. At 50 Hz, 6″ panels at 30% reach 0.518 s — approaching target. At 63 Hz and above, 4″ panels at 30% already achieve target without any TDA treatment.

Coverage has minimal effect in the sub-60 Hz range for panels of any thickness. At 31.5 Hz, increasing coverage from 10% to 60% using 4″ panels changes T60 from 1.426 s to 1.252 s or a 12% improvement from six times the panel area. This confirms that panel absorption is physically limited at these wavelengths regardless of quantity: the mechanism (porous absorption) is not effective in this range. The Nodal 30, by contrast, brings T60 at 28–35 Hz from the 1.0–1.4 s range to 0.38–0.50 s in a typical residential room with four corner units which is a 60–70% reduction that no panel configuration can approach.

The panel depth selector therefore affects the model primarily in the 50–80 Hz transition zone. Switching from 4″ to 6″ panels at 30% coverage reduces T60 at 50 Hz from 0.744 s to 0.518 s which is a meaningful difference in a frequency range where second-order axial modes commonly occur in rooms of 3.5–5 m dimensions. For rooms with dominant 50–70 Hz problems, the panel depth input provides a more accurate prediction of how much treatment remains for the TDA to address. The simulation's Treatment Recommendation panel accounts for this: when 6″ panels are selected, modes in the 50–70 Hz range may already be at target, and the panel correctly removes them from the unit-count projection.

14.6.6 Coverage Sensitivity and Model Limitations

Coverage percentage is user-estimated and constitutes the primary source of uncertainty in the part-treated profile calculation. The model treats coverage as spatially uniform, panels distributed evenly across all six surfaces. In practice, panels are typically concentrated on front and rear walls or specific reflection points, producing a non-uniform distribution. The effective alpha of a non-uniform installation will differ from the area-weighted estimate, particularly in the 50–80 Hz range where the panel contribution is significant but not saturating.

ISO 354 absorption coefficients can exceed 1.0 due to edge effects in the reverberation chamber measurement. The model clips effective alpha at 0.95 through the underlying Eyring-based boundary damping formula rather than explicitly capping OC703 values, which preserves numerical stability without introducing an artificial alpha ceiling. Real-world panel absorption also varies by ±10% from published values depending on mounting method, air gap, frame depth, and room boundary reflection. REW waterfall measurement before and after treatment remains the definitive verification; the dynamic panel model improves the pre-purchase prediction accuracy but does not substitute for measurement.

The PDF report generated by the simulation includes a Broadband Panel Contribution table that exports α_shell, α_panel, α_eff, T60 shell, T60 effective, and panel improvement percentage at each octave band, providing a record of the panel configuration assumed in the prediction for comparison against post-installation REW measurements.

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